EXACT COMPUTER-ASSISTED RESULT

The Greedy Shortest-Superstring Bound for at Most Six Input Words

An Exact Computer-Assisted Proof

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ABSTRACT

Let S be a finite substring antichain of nonempty words over an arbitrary alphabet. The greedy shortest-common-superstring algorithm repeatedly merges an ordered pair of current words having maximum directed suffix-prefix overlap, with ties resolved arbitrarily. We prove that if 1 <= |S| <= 6, then every greedy run returns a common superstring G satisfying |G| <= 2 OPT(S). The proof is exact and computer-assisted. A globally maximal merge is shown to preserve both the substring antichain and all endpoint overlap interfaces, reducing every literal run to an edge-selection chronology on the original labels. After relabeling the final greedy path, 2,880 five-input cases and 86,400 six-input cases remain. Exact rational Farkas certificates prove the factor-two bound in every case; a reverse-and-relabel involution reduces the stored six-input corpus to 43,200 representatives. Verification uses only Python standard-library exact arithmetic. A five-word binary family has legal greedy ratio 2 - 4/(t+4), so the constant is asymptotically sharp already for five inputs. This result does not settle the unrestricted Greedy Superstring Conjecture.

Artifact repository: https://github.com/rishigajjala/greedy-superstring-six-inputs

Scope: arbitrary alphabet and word lengths; one through six reduced input words; every greedy tie resolution.

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1. Introduction

The shortest common superstring problem asks for a shortest word containing every member of a finite input family as a contiguous substring. Tarhio and Ukkonen introduced and analyzed the natural greedy algorithm in 1988: repeatedly merge two current words having a maximum suffix-prefix overlap [2]. They conjectured that every run has approximation ratio at most two. Despite extensive work, the unrestricted conjecture remains open. The strongest published general upper bound for GREEDY is (sqrt(67)+2)/3, approximately 3.396 [3].

This paper concerns a different parameter: the number of input words. Alanko explicitly proposed the five-input case as a finite but nontrivial challenge [1]. We prove that challenge and one further case.

> **THEOREM - Main theorem**
>
> Let Sigma be an arbitrary alphabet and let S be a finite set of nonempty words over Sigma, no one of which is a substring of another. If 1 <= |S| <= 6, then every run of the standard greedy shortest-superstring algorithm, including every resolution of overlap ties, returns a word G such that |G| <= 2 OPT(S).

The parameterization must not be confused with recent work on inputs whose individual words have fixed length k. In particular, Chukhin, Kulikov, Mihajlin, and Smal prove lower bounds for words of length six [6]; their parameter is word length, whereas ours is input cardinality.

Question	Parameter fixed	Status relevant here
Unrestricted Greedy Conjecture	Neither	Open; not settled here
Fixed word length k	Length of each input word	Separate line of work [6]
Present theorem	Number of reduced input words	Ratio at most 2 for 1 through 6

The proof has three conceptual components. First, a structural lemma shows that a globally maximal greedy merge preserves the substring antichain and inherits all external overlaps from its two endpoints. Thus every current word can be represented exactly by a directed path through original inputs. Second, an optimal superstring is equivalent to a maximum-overlap Hamiltonian path through the reduced input family. Third, after canonical relabeling, finitely many chronology/path pairs remain. A necessary-condition linear program is associated with each pair, and exact rational dual certificates prove the desired bound.

The finite calculation is part of the proof. Floating-point optimizer output is used only during discovery and contributes nothing to the theorem. The certificate checkers reconstruct every integer row, check every multiplier sign and stationarity coordinate with fractions.Fraction, and verify the exact bound. Section 8 gives a complete reproducibility ledger.

2. Definitions and conventions

A word is a finite sequence over an arbitrary alphabet Sigma. For words x and y, define the directed overlap

```
ov(x,y) = max { k : suffix_k(x) = prefix_k(y) }.
```

The quantity is directed: ov(x,y) need not equal ov(y,x). For distinct words x and y, their maximum-overlap merge is

```
x merge y = x followed by y[ov(x,y):].
```

An input set is reduced if it is a substring antichain: no member is a substring of another. Standard preprocessing deletes duplicate words and every word contained in another; this does not change OPT. The theorem concerns GREEDY after this preprocessing. It does not identify such a run with an arbitrary run on an unreduced collection.

Starting with a reduced set S, GREEDY repeatedly chooses an ordered pair of distinct current words x,y for which ov(x,y) is globally maximum among all ordered pairs, replaces x,y by x merge y, and stops when one word remains. Every tie may be resolved adversarially.

For an original input s_i, write

```
l_i = |s_i|,      w_ij = ov(s_i,s_j),      L = sum_i l_i.
```

Let OPT(S) denote the minimum length of a common superstring. All words are nonempty, so OPT(S)>0.

COROLLARY - Preprocessing formulation

For an arbitrary finite collection I, first delete duplicates and words contained in other members. If the surviving antichain S has between one and six members, then every subsequent greedy run satisfies |G| <= 2 OPT(I), because OPT(S)=OPT(I).

3. Structural reduction to original-label paths

The next lemma is the essential bridge between literal string merging and the finite overlap model. Global maximality is indispensable.

LEMMA - Endpoint inheritance and antichain preservation

Suppose the current words form a substring antichain. GREEDY chooses A,B with k=ov(A,B) equal to the current global maximum, and let M=A merge B. For every other current word Z,

```
ov(M,Z) = ov(B,Z),          ov(Z,M) = ov(Z,A).
```

Moreover, replacing A,B by M leaves a substring antichain.

Proof. The word M has A as a prefix and B as a suffix, so ov(M,Z) >= ov(B,Z). Put r=ov(M,Z). If r <= |B|, the length-r suffix of M is the length-r suffix of B, and hence r <= ov(B,Z). If r>|B|, then the part of this alignment ending at the end of A has length r-|B|+k>k. It is simultaneously a suffix of A and a prefix of Z, contradicting global maximality. Thus ov(M,Z)=ov(B,Z).

The reverse identity is symmetric. Since A is a prefix of M, ov(Z,M) >= ov(Z,A). If r=ov(Z,M)>|A|, then the portion extending into B gives an overlap from Z to B of length r-|A|+k>k, again contradicting maximality. Otherwise the matched prefix lies inside A, giving equality.

It remains to exclude containment. Because the current family is an antichain, k<|B|, so M is strictly longer than its prefix A. No old word can contain M because it would contain A. Conversely, suppose M contains an old word Z. An occurrence wholly inside the displayed copy of A or B contradicts the old antichain. Any crossing occurrence begins before the displayed copy of B and ends after the end of A. Its prefix through the end of A is then a suffix of A and a prefix of Z of length greater than k, contradicting maximality. Thus no containment is created. QED.

COROLLARY - Path-state invariant

Attach to each current word the ordered list of original labels merged into it. At every step these lists form vertex-disjoint directed paths. If P and Q are two current components, then

```
ov(text(P),text(Q)) = w_(last(P),first(Q)).
```

Consequently every greedy step joins the end of one original-label path to the beginning of another. If the selected original arcs have total weight S_G, then the final greedy length is

```
G = L - S_G.
```

Proof. Induct on the number of merges. The endpoint identities in the lemma preserve every external interface exactly, and the total length drops by precisely the selected overlap at each step. QED.

4. Optimal superstrings and universal overlap inequalities

LEMMA - Hamiltonian-path formula for OPT

For every reduced family S={s_0,...,s_(n-1)},

```
OPT(S) = min_P (L-W(P)) = L-max_P W(P),
W(P)   = sum_(r=0)^(n-2) w_(p_r,p_(r+1)),
```

where P ranges over all permutations (p_0,...,p_(n-1)).

Proof. Choose one occurrence interval of every input in a shortest common superstring. Two chosen intervals cannot be nested, because that would make one input a substring of another. Ordering intervals by their starting positions therefore orders their ending positions strictly as well.

For consecutive intervals of words i,j, let d be the displacement between their starts. If they overlap geometrically, the common block is a suffix of i and a prefix of j, so w_ij >= l_i-d. If they do not overlap, the same inequality is trivial. Thus d >= l_i-w_ij. Summing the displacements and the final word length shows that the superstring has length at least L-W(P) for the induced order P.

Conversely, for any order P, place each next word after the preceding word using their directed overlap. Reducedness makes every such overlap proper, so the right endpoint advances by l_j-w_ij>0. The resulting word contains every input and has length exactly L-W(P). Taking minima proves the formula. QED.

We use three further word inequalities. Their arrow orientations matter.

LEMMA - Directed overlap triangle

For all words A,B,C,

```
ov(A,B)+ov(B,C) <= |B|+ov(A,C).
```

Proof. If ov(A,B)+ov(B,C)-|B| is positive, the two matched portions of B intersect in a block that is simultaneously a suffix of A and a prefix of C. Otherwise the assertion is immediate. QED.

LEMMA - Conditional directed rectangle

Let a=ov(A,B), x=ov(A,C), and z=ov(D,B). If a>=x and a>=z, then

```
ov(D,C) >= max(0,x+z-a).
```

Proof. The length-a alignment A->B identifies prefix_x(C) with the interval [a-x,a] in prefix_a(B). The alignment D->B identifies suffix_z(D) with [0,z]. Their intersection has length max(0,x+z-a) and is simultaneously a suffix of D and a prefix of C. QED.

At a greedy step selecting u->v, every feasible cross edge u->v' and u'->v has weight at most w_uv. Therefore the preceding lemma licenses

```
w_(u,v') + w_(u',v) <= w_(u,v) + w_(u',v').
```

LEMMA - Pair in an optimum

If all inputs occur in a word of length O, then for every pair i,j,

```
w_ij+w_ji >= l_i+l_j-O.
```

Proof. Order chosen occurrences by their starts. If i begins before j at displacement d, then d+l_j<=O. When the occurrences intersect, w_ij>=l_i-d>=l_i+l_j-O; when they do not, the right side is nonpositive. Adding w_ji>=0 proves the assertion. The other start order is symmetric. QED.

5. Canonical cases and the necessary-condition LP

Fix n in {5,6}. By the path-state invariant, the final greedy word determines a Hamiltonian path through the original labels. Relabel that final path as

```
0 -> 1 -> ... -> n-1.
```

Its canonical edges are e_i=(i,i+1), 0<=i<n-1, but GREEDY may have selected them in any chronological order sigma, a permutation of the n-1 edges. After a prefix of sigma, the chosen edges form a path forest whose components are exactly the joined contiguous blocks of the canonical final path.

A directed edge (u,v) is feasible at that stage precisely when u has no selected outgoing edge, v has no selected incoming edge, and u,v lie in different weak components of the selected forest. This is exactly the set of current literal merges by endpoint inheritance.

Let P be a nominated optimal Hamiltonian path. Normalize OPT=1 and introduce n^2 variables

```
l_i = |s_i|/OPT,          w_ij = ov(s_i,s_j)/OPT   (i!=j).
```

For each pair (sigma,P), define Q_(sigma,P)^(n) by the following constraints.

1. Nonnegativity and endpoint caps:

```
0 <= l_i <= 1,       0 <= w_ij <= l_i,       0 <= w_ij <= l_j.
```

2. Every directed triangle, for distinct i,j,k:

```
w_ij+w_jk <= l_j+w_ik.
```

3. Every pair-in-an-optimum row:

```
l_i+l_j-w_ij-w_ji <= 1.
```

4. At each chronology step, if e is selected and f is any feasible edge,

```
w_e >= w_f.
```

5. At each chronology step selecting u->v, every conditional rectangle for feasible cross edges u->v' and u'->v:

```
w_(u,v')+w_(u',v) <= w_(u,v)+w_(u',v').
```

6. The nominated optimal-path equality:

```
sum_i l_i - sum_(ij consecutive in P) w_ij = 1.
```

The objective is the normalized greedy output length

```
G = sum_i l_i - sum_(i=0)^(n-2) w_(i,i+1).
```

> **PROPOSITION - Soundness of the relaxation**
>
> Every literal greedy run on n reduced inputs, together with any optimal Hamiltonian path supplied by the OPT formula, gives a point of exactly one Q_(sigma,P)^(n), and its LP objective is |G|/OPT.

Proof. Relabel the final original-label path canonically and record its actual selection chronology. Endpoint inheritance identifies current merges with the feasible-edge predicate, so global greedy maximality gives every dominance row and licenses every included rectangle. The word lemmas give endpoint, triangle, and pair rows. The optimal Hamiltonian path gives the equality, and telescoping gives the objective. QED.

The polyhedron is deliberately a relaxation. It neither asserts that every feasible overlap matrix is realizable by words nor requires P to dominate every other Hamiltonian path. Omitting such conditions enlarges the feasible set, so an upper bound on Q_(sigma,P)^(n) remains valid for literal strings.

6. Exact dual certificates

Write the minimization form of a case as

```
minimize c^T x = -G
subject to A x <= b,      e^T x = 1,
           x >= 0,        l_i <= 1.
```

The upper bounds l_i<=1 are kept separate below.

LEMMA - Exact certificate criterion

Suppose there are rational multipliers y<=0, a free rational z, lower-bound multipliers q>=0, and length-upper-bound multipliers r<=0 such that

```
A^T y + e z + q + r = c
```

coordinatewise, and

```
b^T y + z + sum_i r_i >= -2.
```

Then every feasible point satisfies G<=2.

Proof. For any feasible x,

```
c^T x = y^T A x + z e^T x + q^T x + r^T l
        >= y^T b + z + sum_i r_i
        >= -2.
```

The first inequality uses y<=0 with Ax<=b, q>=0 with x>=0, and r<=0 with l_i<=1. Since c^T x=-G, the conclusion follows. QED.

All stored multipliers are exact rationals. The verifiers reject an incorrect sign, duplicate or out-of-range sparse index, stationarity mismatch, forged bound, missing case, extra record, or unexpected metadata.

6.1. Five inputs

There are 4! possible greedy chronologies and 5! nominated paths, hence

```
4! * 5! = 2,880
```

cases. The exact corpus contains one certificate for every case.

THEOREM - Five-input finite certificate theorem

Every point in every Q_(sigma,P)^(5) satisfies G<=2.

The standard-library verifier confirms all 2,880 certificates. The weakest dual lower bound is exactly -2 and the largest certificate support is 20.

6.2. Six inputs and the symmetry involution

There are

```
5! * 6! = 86,400
```

raw chronology/path pairs. Define r(i)=5-i and transform variables by

```
l'_i  = l_(r(i)),
w'_ij = w_(r(j),r(i)).
```

This reverses every directed edge and reflects the canonical labels. It maps a canonical edge (i,i+1) to (4-i,5-i), and maps a path (p_0,...,p_5) to

```
(r(p_5),...,r(p_0)).
```

Endpoint caps, directed triangles, pair rows, exact feasible-edge sets, licensed rectangles, the optimum equality, and the objective are invariant. The map is an involution. Only the middle canonical edge is fixed, so no ordered list containing all five distinct canonical edges can be fixed entrywise. Thus the 120 chronologies split into 60 two-element orbits.

THEOREM - Six-input finite certificate theorem

Every point in every Q_(sigma,P)^(6) satisfies G<=2.

The corpus stores certificates for 60*720=43,200 representatives. The strengthened verifier checks the variable permutation, edge/order/path involutivity, disjoint orbit representatives, complete coverage of all 120 chronologies, exact row-with-right-hand-side multiset invariance for all 60 order pairs, objective invariance, all 43,200 transformed path equalities, every metadata field, and every exact dual certificate. The weakest bound is -2 and the largest certificate support is 29.

THEOREM - Main theorem, completed

For every reduced input family S with 1<=|S|<=6, every greedy run satisfies |G|<=2 OPT(S).

Proof. The cases |S|<=4 are proved without computation in Appendix A. For five or six inputs, map the actual run and an optimal path into the corresponding normalized relaxation by the soundness proposition. The exact finite certificate theorem for that cardinality gives G<=2, and rescaling yields |G|<=2 OPT(S). QED.

7. Sharpness at five inputs

PROPOSITION - Asymptotic sharpness

For every integer t>=3, the five binary words

```
A = 001,          B = 0 1^(t-1),
C = 101,          D = 1^(t-1) 0,
E = 1^t
```

form a reduced family with a legal greedy run of length 2t+4 and optimum t+4. Hence the ratio is

```
(2t+4)/(t+4) = 2 - 4/(t+4),
```

which tends to two.

Proof. The five words are distinct and no shorter word occurs in a longer one, so the family is reduced. Consider the run

```
B -> D          overlap t-1,
C -> (B merge D) overlap 2,
A -> ...         overlap 1,
E -> ...         overlap 0.
```

At the first step no proper overlap can exceed t-1, and B->D attains it. After this merge, C and A each have overlap two into the composite and no larger overlap exists. After choosing C, every positive remaining overlap has length one. After choosing A, both directions involving E have overlap zero. Thus every displayed step is globally greedy, including all ties.

The total input length is 3t+6 and the run saves (t-1)+2+1=t+2, so its output has length 2t+4. The order A,B,E,D,C saves

```
2 + (t-1) + (t-1) + 2 = 2t+2
```

and therefore gives a superstring of length t+4. This is optimal. If a Hamiltonian path uses two of the three weight-(t-1) arcs B->E, B->D, E->D, its other two edges have weight at most two, giving total saving at most 2(t-1)+4. If it uses at most one long arc, its saving is at most (t-1)+6<=2t+2 for t>=3; with no long arc it is at most 8<=2t+2. Thus no path saves more than 2t+2. QED.

COROLLARY - Exact fixed-cardinality constant

The supremum of the worst greedy ratio over reduced instances with at most six inputs is exactly two. The lower bound already occurs as a limit over instances with exactly five inputs.

8. Reproducibility and integrity ledger

The certificate corpora are part of the proof. The repository contains the integer model builders, deterministic generators, exact verifiers, corpora, manuscript source, and audit notes. No commercial solver or third-party Python package is used during verification.

Inputs	Raw cases covered	Stored certificates	Corpus bytes	SHA-256
5	2,880	2,880	2,221,399	e733e5cb361e515db6fe257789f3991b f5f264a3f241814d9a74464439b0fe66
6	86,400	43,200	897,380	53ef042c6c7b4c59ca2beac9b960fcd9 9ea522956d3212efdbd122a35119ab01

From the repository root, run

```
python3 verify_five_string_certificates.py five_string_certificates.json

python3 generalize/verify_six_string_certificates_v2.py \
  generalize/six_string_certificates.jsonl.gz

python3 verify_sharp_family.py --max-t 250
```

The expected exact summaries are

```
five: verified_cases=2880,
      weakest_dual_lower_bound=-2,
      maximum_certificate_support=20

six: verified_representative_cases=43200,
     covered_cases_by_involution=86400,
     weakest_dual_lower_bound=-2,
     maximum_certificate_support=29

sharpness: t=3..250 verified
```

The six-input records are positional rather than carrying explicit case identifiers. This is an auditability limitation, not a coverage gap: the verifier deterministically regenerates the expected chronology/path order and tests each record against its corresponding case. It rejects missing or trailing records.

An internal adversarial audit independently rederived the bridge lemmas, feasible-edge predicate, weak-duality signs, symmetry, and case counts. It also performed exhaustive small-word tests, cross-implementation comparisons, random literal-run embeddings, and negative controls that deliberately corrupted signs, values, bounds, row coefficients, case order, stream length, and metadata. These checks are internal validation, not independent peer review.

9. Scope and limitations

This manuscript proves a fixed-input-cardinality theorem only. It does not prove or disprove the unrestricted Greedy Superstring Conjecture, and it establishes neither a seven-input theorem nor a seven-input counterexample.

The necessary-condition LP used here is sufficient through six inputs but admits non-word-realizable points above ratio two at seven inputs. This is evidence that pairwise overlaps, triangles, rectangles, and pair-in-an-optimum inequalities do not by themselves characterize global word realizability. Separate occurrence-alignment and periodicity lemmas in the research notes eliminate the saved abstract obstructions, but they do not classify every seven-input case and are not used in the theorem proved here.

The result is a new computer-assisted research artifact and has not been externally peer reviewed. Before journal submission, human authors should check every mathematical argument, fix authorship and licensing, archive the exact corpora and source at an immutable repository, and take responsibility for the claims.

10. AI-use disclosure

This draft, its exact-search code, certificate-generation workflow, and internal adversarial audits were developed with substantial assistance from OpenAI Codex and multiple model instances. The final theorem is supported by explicit mathematical reductions and independently checkable rational certificates, but AI assistance is not a substitute for external peer review. Any submitting authors must review the manuscript and proof artifacts in full and accept responsibility under the target venue's policy.

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Appendix A. The cases of at most four reduced inputs

The case m=1 is immediate. Suppose 2<=m<=3, and let a be the first maximum overlap selected by GREEDY. Every edge of an optimal Hamiltonian path has weight at most a, so

```
L = OPT+W(P*) <= OPT+(m-1)a.
```

GREEDY saves at least a. Therefore

```
G <= OPT+(m-2)a <= 2 OPT,
```

because a is at most the length of an input word and hence at most OPT.

Now let m=4, and denote the three optimal-path overlaps by p,q,r in order. Endpoint caps give

```
L >= p+r+max(p,q)+max(q,r),
```

and consequently

```
2(p+q+r)-L <= min(p,q)+min(q,r).                 (A.1)
```

Let a,b be the first two greedy overlaps. If at least one optimal edge remains feasible after the first merge, then

```
b >= min(p,q,r),
a >= max(min(p,q),min(q,r)).
```

Since min(p,q,r)=min(min(p,q),min(q,r)), their sum is at least the right side of (A.1), proving the desired bound.

The only directed merge that can make all three edges of a four-vertex optimal path infeasible is the reverse middle edge. Relabel the optimum as A->B->C->D and suppose GREEDY selects C->B with weight a. Directed triangles leave feasible edges

```
w_AC >= p+q-|B|,         w_BD >= q+r-|C|.
```

Thus

```
b >= max(0,p+q-|B|,q+r-|C|).
```

Put x=p+q-|B| and y=q+r-|C|. Endpoint caps imply x<=p<=a and y<=r<=a, so

```
x+y <= a+max(0,x,y) <= a+b.
```

Finally,

```
2(p+q+r)-L-a-b
 <= p+2q+r-|B|-|C|-a-b
  = x+y-a-b
 <= 0.
```

Hence G<=2 OPT for four inputs as well.

Appendix B. Exact row schema and deterministic case order

For fixed n, the canonical greedy edges are stored as

```
E = ((0,1),(1,2),...,(n-2,n-1)).
```

Chronologies are lexicographically ordered permutations of E. Optimal paths are lexicographically ordered permutations of range(n). For a selected chronology prefix, feasibility is determined by the following mathematical predicate.

```
feasible(u,v,chosen):
    reject if u=v
    reject if u already has a chosen outgoing edge
    reject if v already has a chosen incoming edge
    reject if u and v are in the same weak component of chosen
    otherwise accept
```

Rows are emitted in a fixed family order: endpoint caps, directed triangles, pair rows, then stepwise greedy-dominance and licensed rectangle rows. The optimum equality and objective are generated separately. Integer coefficients are used throughout; normalization enters only through right-hand sides equal to zero or one.

For n=6 the stored order first selects the 60 lexicographically chosen representatives of the reverse-and-relabel chronology orbits, then all 720 nominated paths. The v2 verifier recomputes the orbit and equality maps rather than trusting the header.

Appendix C. Certificate representation and checker logic

Each sparse certificate stores the claimed bound, nonzero inequality multipliers y, nonzero lower multipliers q, nonzero length-upper multipliers r, and the free equality multiplier z. Rational numbers are encoded as strings accepted by fractions.Fraction. The checker rejects duplicate sparse indices and verifies the expected sign before any arithmetic.

For each case it rebuilds A,b,e,c and performs the following exact checks.

```
1. Decode every multiplier as a rational number.
2. Check y<=0, q>=0, r<=0 and all sparse index ranges.
3. Accumulate A^T y + e z + q + r coordinatewise.
4. Require exact equality with c in every coordinate.
5. Compute beta = b^T y + z + sum_i r_i exactly.
6. Require beta to equal the stored bound and beta>=-2.
7. Reject a missing certificate or any trailing record.
```

For six inputs, the checker additionally constructs the 36-coordinate involutive variable permutation, compares complete row/right-hand-side multisets, compares objectives, verifies all transformed equalities, and checks exact orbit coverage.

Appendix D. Internal audit summary

The internal hostile audit included the following independent checks.

- All path-forest subsets agreed with an independently written Hamiltonian-extension enumerator.
- Every raw merge history reduced to the expected canonical chronology multiplicity.
- Small-word exhaustions found no violation of endpoint inheritance, antichain preservation, directed triangles, or conditional rectangles.
- Literal optimal-superstring enumeration agreed with the Hamiltonian-path formula on exhaustive small antichains.
- Random five- and six-word literal greedy runs mapped into their nominated LP cases without a failed row, including many zero-overlap selections.
- The exploratory floating and exact integer model builders agreed row for row.
- Deliberately corrupted certificates were rejected for bad signs, changed values, forged bounds, changed row coefficients, missing records, swapped case order, truncated streams, and false metadata.

The five-input review found one expository omission: the preprocessing corollary originally invoked a first overlap when only one word survived. The statement was repaired by separating m=1 before the m=2,3 argument. No theorem-level defect was found in the final package.